MODULE

'I'

Definition of Artificial Intelligence

Artificial Intelligence or AI is a branch of computer science focused on building systems and machines that can performed tasks typically requiring human intelligence. These tasks includes learning, reasoning, problem solving, perception, language understanding and decision making.

Examples of Artificial Intelligence

1. **Voice Assistants:**
   They used tools like Natural Language Processing (NLP) to understand commands and respond accordingly.

   For example: Alexa, Siri, Google Assistant, etc.

2. **Recommendation Systems:**
   AI analyses user past activity like shopping and viewing history to suggest relevant products and content that user may like.

   For example: Amozon, Netflix, Youtube, etc.

3. Self-driving Cars:
Companies like Tesla uses AI for object detection, decision making and navigating roads.

For example; Waymo, Pony.ai, etc.

4. Chatbots & Customer Service:
These handles routine customer query using Machine Learning (ML) and NLP.

For example; Customer Support Chatbots.

5. Healthcare Diagnosis:
AI models can analyse X-rays and other medical data/report to detect diseases like cancer with high accuracy.

For example; iCAD, Google Health's Retinal Screening, etc.

AI Based Mechanical Engineering System (MES)

An AI based MES integrates artificial intelligence techniques into mechanical processes or systems to improve performance, automate decision making, reduce human errors and optimize design or operation.

These systems often use sensors, data analysis, machine learning and predictive modelling to enhance traditional mechanical engineering task.

Mechanical engineering involves designing, analyzing and manufacturing mechanical systems such as: engines, machines, HVAC Systems (Heating, Ventilation and Air Conditioning) and manufacturing equipment. When AI is introduced, it enables:

i. Predictive Maintenance of machines.
ii. Automated fault detection.
iii. Intelligent design optimization.
iv. Adaptive control system.
v. Robotics & Automation in manufacturing.

AI learns from historical data, sensor inputs and operational feedback to make intelligent decisions, improving efficiency, safety and life-span of mechanical systems.

Examples of AI in Mechanical Engineering System

1. **Predictive Maintenance:**
Using AI to predict machine failure before they happen. AI analyses vibrations, temperature and pressure data to forecast breakdowns.

2. **Smart HVAC System:**
Adaptive temperature and airflow control. AI adjusts HVAC operation based on occupancy and weather predictions.

3. **Design Optimization:**
Generative design and Topology optimization. AI creates optimized design based on constraints and performance goals.

4. **Industrial Robots:**
Robots with AI vision and decision making. Used for ~~assembly,~~ assembling, welding and material handling in smart factories.

5. **Additive Manufacturing (3D Printing):**
AI controlled print parameters. Enhances part quality and reduces waste during printing.

Types of AI Based Mechanical Engineering System

1. Predictive Maintenance System
2. Intelligent Design Optimization System
3. Smart Manufacturing System
4. Adaptive Control System
5. Autonomous Mechanical Systems [Robotics]
6. Fault Detection & Diagnostics System.

1. Predictive Maintenance System:

AI analysis sensor and operational data to predict when a mechanical component is likely to fail. This allows pro-active maintenance, reducing unplanned down time.

Example: Monitoring temperature and vibrations of motors to detect bearing wear.

Technology Involved: Machine learning, IoT sensors, and Data analytics.

Case Study:

Case: Siemens Gas Turbine Monitoring

Siemens uses AI for real-time monitoring of gas turbine. The system credits failure upto weeks in advanced, reducing down time by 30% and saving in millions in service cost.

2. Intelligent Design Optimization System:

AI tools automate and optimized mechanical components for system design, improving efficiency, reducing weight and lowering cost while needing all constraints.

Example: Using Generative design to create light-weight structural components for aircraft.

Technology Involved: Genetic algorithm, Neural network, topology optimization.

Case Study:

Case: Auto-desk and Air-bus

Air-bus used auto-desk AI powered generative design to create a new partition for its A320 aircraft. The result was a structured 45% lighter, saving fuel and reducing emissions.

3. <u>Smart Manufacturing System</u>: [Industry 4.0]

AI integrated into manufacturing systems enables real-time decision making, process control and adaptive automation.

<u>Example</u>: Robotic-arms using computer vision to assemble complex products.

<u>Technology Involved</u>: AI, Robotics, Digital twins, Cyber Physical Systems.

<u>Case Study</u>:

<u>Case: BMW Smart Factory</u>

BMW uses AI-driven robots and digital twins for assembly line management. AI adapts in real time to changes in-parts or materials, increasing production flexibility and speed.

4. <u>Adaptive Control System</u>:

AI is used to create control systems that can ~~confr~~ learn from feedback and adjust operation automatically.

<u>Example</u>: HVAC systems that learn building usage patterns and adjust temperature control.

<u>Technology Involved</u>: Reinforcement learning, fuzzy logic, real-time data feedback.

<u>Case Study</u>:

<u>Case: AI-based HVAC Optimization at MS Campus</u>

Microsoft implemented AI control in their HVAC system. It reduces energy consumption by 25% ~~by~~ while maintaining occupant comfort by continuously adjusting based on room usage and weather.

5. Autonomous Mechanical Systems:

AI enables robots and mechanical systems to make decisions, move and interact with their environment without constant human control.

Example: Self-navigating delivery robots in warehouses.

Technology Involved: Deep learning, Computer vision, SLAM (Simultaneous Localization & Mapping).

Case Study:

Case: Amazon's Robots

Amazon uses AI-powered robots to move shelves of product to human workers. AI coordinates thousands of robots, reducing order processing time by 50%.

6. Fault Detection & Diagnostics System:

AI systems detect anomalies in mechanical system and identify the root-cause of faults quickly, minimizing damage.

Example: Detecting miss-alignment or imbalance in rotating machinery using vibration analysis.

Technology Involved: Anomaly detection, Supervised learning, Signal processing.

Case Study:

Case: Rolls-Royce Engine Health Monitoring

Rolls-Royce uses AI to ~~maintain~~ monitor aircraft engine data in real-time. The system detect early sign of component fatigue or failure, improving safety and scheduling pro-active repairs.

	Types of AI-system	Key Benefit	Example	Case-Study
1.	Predictive Maintenance	Reduced Downtime	Vibration based motor-monitoring	Siemens Turbine.
2.	Design Optimization	Light-weight, cost effective	Generative Part Design	Airbus A320 Partition.
3.	Smart Manufacturing	Efficient Production	Adaptive Robotic Arms	BMW Smart Factory.
4.	Adaptive Control	Energy Savings	Smart HVAC System	Microsoft Campus HVAC.
5.	Autonomous System	Automation	Ware-house Robots	Amazon Robotics.
6.	Fault Detection	Early Diagnosis	Air Vibration Monitoring	Rolls-Royce Engine.

Conclusion: AI is transforming mechanical engineering by making systems smarter, more adaptive and data-driven. From intelligent robots to self optimizing machine, AI-based mechanical engineering system are more efficient and automated future in engineering.

Machine Learning (ML)

Machine Learning is a sub-field of AI that enables computers to learn from data and make decisions for prediction without being explicitly programmed for every task.

At its core, machine learning is about training algorithms to identify patterns in data and used those patterns to predict outcomes, classify information, make decisions and automate tasks.

Rather than following static rules, ML models learn from experience - the more data they process, the better they become.

How it works?

1. Data Collection:
   Gather input data (like sales number, images, sensored data).

2. Model Training:
   Feed data into an algorithm to find patterns or rules.

3. Testing:
   Validate the model with new/unseen data.

4. Prediction / Decision:

Apply the model to real world scenarios.

Types of Machine Learning